

Computer Science & IT

Database Management System



Relational Model & Normal Forms

Lecture No. 11



By- Vishal Sir



Recap of Previous Lecture



- ✓ Topic Lossless join decomposition
- ✓ Topic Projection, Selection, and Cross product
- ✓ Topic Natural join
- ✓ Topic Lossless Natural Join

Topics to be Covered



Topic

Normal forms



Topic

First normal form (1NF)



Topic

Redundancy in relation because of FD





Lossless Natural Join.

Note Consider the following relational table.

R

A	B	C
1	1	1
2	1	2
3	2	1

① If relation R is decomposed into two subrelations $R_1(AB)$ & $R_2(BC)$, then check whether the decomposition is lossless Join decomposition or not?

$R_1 =$

A	B
1	1
2	1
3	2

$R_2 =$

B	C
1	1
1	2
2	1

$R_1 \bowtie R_2 =$

A	B	C
1	1	1
1	1	2
2	1	1
2	1	2
3	2	1

tuples from R

Extra tuples {Spurious tuples}
Here,

$$R_1 \bowtie R_2 \supset R$$

∴ Lossy Join decomposition

- * Common attribute b/w R_1 & R_2 is B
 - * Values of B are neither unique in R_1 nor unique in R_2
 - * B is neither a S.K of R_1 nor a S.K of R_2
- Hence Lossy Join decomposition

Note Consider the following relational table.

R

A	B	C
1	1	1
2	1	2
3	2	1

② If relation R is decomposed into two subrelations $R_1(AB)$ & $R_2(AC)$, then check whether the decomposition is lossless Join decomposition or not?

$R_1 =$

A	B
1	1
2	1
3	2

$R_2 =$

A	C
1	1
2	2
3	1

$R_1 \bowtie R_2 =$

A	B	C
1	1	1
2	1	2
3	2	1

- Common attribute b/w R_1 & R_2 is A.
- Values of A are unique in R_1 as well as in R_2
- i.e. A is a Key in R_1 as well as in R_2
∴ Lossless Join decomposition

Here, $R_1 \bowtie R_2 = R$
∴ Lossless Join decomposition

Note Consider the following relational table.

R

A	B	C
1	1	2
2	1	2
3	2	1

③ If relation R is decomposed into two subrelations $R_1(AB)$ & $R_2(BC)$, then check whether the decomposition is lossless Join decomposition or not?

$R_1 =$

A	B
1	1
2	1
3	2

$R_2 =$

B	C
1	2
2	1

$R_1 \bowtie R_2 =$

A	B	C
1	1	2
2	1	2
3	2	1

- Common attribute is B.
- Values of B are not unique in R_1 but they are unique in R_2 .
- i.e., B is not a SK in R_1 , but B is a SK in R_2 .

And decomposition is lossless Join decomposition.

Here, $R_1 \bowtie R_2 = R$
∴ Lossless Join decomposition



Topic : NOTE

- * If relation R is decomposed into two sub-relations R_1 and R_2 then decomposition is lossless Join decomposition if and only if
- ① Attributes of $R_1 \cup$ Attributes of $R_2 = R$
 - and ② Attributes of $R_1 \cap$ Attributes of $R_2 \neq \emptyset$ { i.e. there must be some common attributes }
 - and ③ (Attributes of $R_1 \cap$ Attributes of R_2) must be a Super key of either R_1 or R_2 or both ($R_1 \& R_2$)

i.e. Common attributes of $R_1 \& R_2$

H.W.



#Q. Let $R(A, B, C, D, E)$ be the relational schema with following FD set

$$F = \{ AB \rightarrow C, C \rightarrow D, B \rightarrow E \}$$

Which of the following decomposition is/are lossless join decomposition?

(i) $\{R_1(\overline{A}\overline{B}\overline{C}), R_2(\overline{C}\overline{D})\}$

Attributes of $R_1 \cup$ Attributes of $R_2 \neq$ Attributes of R
 $\therefore$ Lossy

(ii) $\{R_1(ABC), R_2(DE)\}$

$R_1 \cup R_2 = R \checkmark$
 $\{A, B, C\} \cap \{D, E\} = \emptyset \therefore$ Lossy.

(iii) $\{R_1(ABC), R_2(CDE)\}$

$$R_1 \cup R_2 = R \checkmark$$

$\{A, B, C\} \cap \{C, D, E\} = \{C\} \neq \emptyset \checkmark$
 $(C)^+ = \{C, D\}$
 $\therefore$ Neither a S.K. of R_1 nor a S.K. of R_2
 $\therefore$ Lossy

☒ (iv) $\{R_1(ABCD), R_2(BE)\}$

$$R_1 \cup R_2 = R \checkmark$$
$$\{A, B, C, D\} \cap \{B, E\} = \{B\} \neq \emptyset \checkmark$$

$(B)^+ = \{B, E\}$
All attributes of $R_2 \therefore B$ is S.K. of $R_2 \checkmark$



Topic : NOTE

$$(R_1 \bowtie R_3) \bowtie R_1$$



① $R_1 \bowtie R_2 = R_2 \bowtie R_1$ { i.e., Natural join is Commutative }
{ tuple wise we will get same tuples }

② $(R_1 \bowtie R_2) \bowtie R_3 = R_1 \bowtie (R_2 \bowtie R_3)$
{ i.e., Natural join is associative }

#Q. Let $R(A, B, C, D, E, F)$ be the relational schema with following FD set
 $F = \{ AB \rightarrow C, BC \rightarrow A, AC \rightarrow B, AD \rightarrow E, B \rightarrow D, E \rightarrow F \}$

Which of the following decomposition is lossless join decomposition.

(i) $\{R_1(ABC), R_2(ABDE), R_3(EF)\}$

(ii) $\{R_1(ABC), R_2(ADF), R_3(ACDE)\}$

(iii) $\{R_1(AB), R_2(BC), R_3(ABDE), R_4(EF)\}$

(i) $R_1(ABC)$, $R_2(A,B,D,E)$, $R_3(EF)$

$\eta = AB$

$(AB)^+ = \{A, B, C, D, E, F\}$

ie. (AB) is S.K. of
 R_1 as well as R_2
 $\therefore$ We can join R_1 & R_2

$R_1 \bowtie R_2 = (A, B, C, D, E)$

$\eta = \{E\}$

$(E)^+ = \{E, F\}$

'E' is the S.K. of R_3
 $\therefore$ We can join $(R_1 \bowtie R_2)$ with R_3

$(R_1 \bowtie R_2) \bowtie R_3 = (A, B, C, D, E, F)$

Join is lossless at
Every point of join.
 $\therefore$ overall decomposition is
Lossless Join decomposition

(ii) $R_1(ABC)$ $R_2(ADF)$ $R_3(ACDE)$

$$n = A$$

$$(A)^+ = \{A\}$$

A is not a S.K.
of either relation
 $\therefore$ Can not be joined

$$n = AC$$

$$(AC)^+ = \{A, C, B, D, E, F\}$$

(AC) is S.K. of both the relations.
 $\therefore$ We can join R_1 & R_3

$$R_1 \bowtie R_3 = (A, B, C, D, E)$$

$$n = AD$$

$$(AD)^+ = \{A, D, E, F\}$$

(AD) is S.K. of R_2

$\therefore$ We can join $(R_1 \bowtie R_3)$ with R_2

$$(R_1 \bowtie R_3) \bowtie R_2 = (A, B, C, D, E, F)$$

There exists an order in which
relations can be joined such that
join is lossless at every point of join
 $\therefore$ Overall decomposition is
lossless Join decomposition

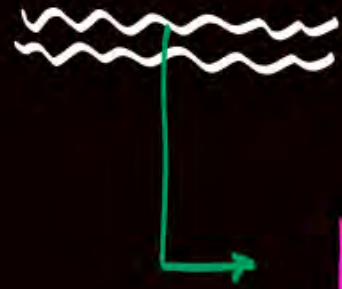
(iii)

$R_1(AB)$

$R_2(BC)$

$R_3(ABDE)$

$R_4(EF)$



We will not be able to join (BC) with other relations in any order such that the join is lossless, so overall decomposition is loss Join decomposition



Topic : Normalization



Normalization is the process of reducing/eliminating the redundancy present in the relation.



Topic : Normal forms



There are various normal forms

1 NF, 2 NF, 3 NF, BCNF
(Boyce Codd Normal Form)

4 NF

- Upto BCNF we will try to eliminate the redundancy present in the relation because of functional dependency.
- If the relation is in BCNF, then there will be 0% redundancy in that relation because of functional dependency but redundancy may be present in that relation because of Multi-valued dependency.

- 4th normal form is related to Multi-valued dependency.
- In 4NF we will try to eliminate the redundancy present in the relation because of Multi-valued dependency.



Topic : Normal forms





Topic : First normal form (1NF)

A database is in 1NF only if it does not contain any multi-valued attributes.

it is multi-valued attribute

Sid	Proj-no.
S ₁	{ P ₁ , P ₂ }
S ₂	{ P ₂ , P ₃ }
S ₃	P ₁

Bring the DB into 1NF

{ i.e., Convert multi-valued attribute into single valued attribute }

Multi value attribute is present in database, so it is not in 1NF

Sid	Proj-no.
S ₁	P ₁
S ₁	P ₂
S ₂	P ₂
S ₂	P ₃
S ₃	P ₁

Both are single valued attributes

No multi-valued attribute is present in the database, so it is in 1NF

★ By default normal form a relation is "1NF"
i.e, Every relation is at least in 1NF

1NF does not eliminate redundancy,
it only converts multi-valued attributes
into single valued attributes.



Topic : Redundancy in relation because of FD

1st Rule : A non-trivial functional dependency $X \rightarrow Y$ does not cause any redundancy in the relation if "X" is a Super Key of that relation.



Topic : Redundancy in relation because of FD

2nd Rule : A non-trivial functional dependency $X \rightarrow Y$ will cause redundancy in the relation if "X" is not a super key of that relation

$A \rightarrow BC$

|||

$A \rightarrow B$

$A \rightarrow C$

A is super key
there is

no redundancy

w.r.t $A \rightarrow B$

$A \rightarrow C$

A	B	C
1	1	4
2	2	7
3	1	4
4	1	4
5	2	7
6	3	7

$B \rightarrow C$

and 'B' is not a s.k of relation.

∴ There is redundancy
in the relation

w.r.t. $B \rightarrow C$

Note:- In non-trivial functional dependency $X \rightarrow Y$, if "X" is not a super key of that relation, then "Y" can never be the superkey of that relation.

∴ A non-trivial functional dependency $X \rightarrow Y$ in which none of X or Y is a Super Key will always cause redundancy in the relation.

H.W

write

→ Possible types of non-trivial functional dependencies that may cause redundancy in the relation



THANK - YOU